

Research Proposal: GenAI as Process Innovation

Title: The Impact of Multi-Layered AI Infrastructure on Organizational Performance Mediated by Trust and Knowledge Sharing: An Empirical Study of "Chat-to-Work" Frameworks

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Document Type:

Independent Research Blueprint (Academic Extension for FinCUI-Trust Framework)

1. Introduction & Problem Statement The financial services industry is currently navigating a fundamental paradigm shift from "Deterministic SOPs" (Standard Operating Procedures), where staff follow rigid rules, to "Probabilistic Workflows," where employees must collaborate with AI agents that exercise autonomous judgment. While industry reports often highlight Generative AI as a simple efficiency tool, real-world deployments reveal a significant "Performance Gap". This research argues that AI performance is not merely a product of algorithmic power, but of Governance Architecture. The study focuses on the implementation of a comprehensive Internal GenAI Platform—a suite of tools designed to make AI accessible and governed—and how this infrastructure facilitates the "Chat-to-Work" transition.

2. Theoretical Framework: Integrating TOE and Process Innovation

To model the mechanics of this transition, this study employs the Technology-Organization-Environment (TOE) framework, mapping specific platform components to key technological and organizational variables:

- Technological Context (ARouter & Explainability): The ARouter (LLM Gateway) serves as the central governing node, routing requests to various models while tracking token usage and maintaining security. This research evaluates how the transparency and control provided by a gateway influence "Trust-in-Process" — the willingness of employees to rely on probabilistic outputs.
- Organizational Context (ASkills & Structural Agility): The ASkills (Agent Skill Hub) acts as a central registry for reusable AI capabilities. By providing standardized, security-scanned "skills" for teams to publish and consume, the platform enhances "Structural Agility," allowing the organization to pivot work processes rapidly.
- Environmental Context (Regulatory Pressure): In highly regulated FinTech environments, the platform's built-in audit trails and compliance guardrails provide the "Institutional Trust" necessary for large-scale adoption.

3. Mediating Mechanisms: Knowledge Sharing via Knowledge Hub and Operation Asst

A critical hypothesis of this study is that AI infrastructure serves as a catalyst for "Knowledge Sharing Behavior".

Knowledge Hub: As a company-wide knowledge hub with web-connected AI, the Knowledge Hub lowers the cost of information retrieval and synthesis. The research examines if such a hub encourages staff to contribute their expertise back into the digital

ecosystem. Operation Asst: By embedding AI directly into specific operational workflows, tools like the Operation Asst normalize AI-human collaboration, potentially reducing the cognitive load and "Adoption Friction" often associated with new technology.

4. Research Objectives & Methodology

The primary objective is to quantify how this infrastructure translates into "Operational Performance" (Efficiency and Quality). Specifically, this study aims to: Verify the impact of ARouter's governance transparency on employee "Trust-in-Process". Analyze how the modularity of ASkills drives "Process Innovativeness" — the ability of teams to reinvent their own daily workflows. Evaluate the role of the Knowledge Hub in fostering a culture of automated knowledge sharing. Utilize Quantitative PLS-SEM (Partial Least Squares Structural Equation Modeling) to triangulate perceptual data from staff with objective engineering logs (e.g., measuring deltas in task completion time and error reduction).

5. Significance & Contribution

This research provides an empirical blueprint for "FinTech AI Governance". It moves beyond surface-level adoption metrics to show how a multi-layered infrastructure — incorporating gateways, skill hubs, and knowledge bases—can systematically eliminate organizational resistance. The findings will offer a scalable "Human-AI Symbiosis" protocol for global financial institutions seeking to deploy GenAI within zero-tolerance, high-stakes environments.